Q. 1 (Probability gymnastics)**.** Let A, B, C be any events.

a. Show that if $A \subseteq B$, then $\mathbf{P}(B \backslash A) = \mathbf{P}(B) - \mathbf{P}(A)$.

b. Show that $\mathbf{P}(A \cup B) \leq \mathbf{P}(A) + \mathbf{P}(B)$.

c. Show that $\mathbf{P}(A \cap B \cap C) = \mathbf{P}(A|B \cap C)\,\mathbf{P}(B|C)\,\mathbf{P}(C)$.

Q. 2 (Sherlock Holmes)**.** An important woman has a meeting in her office. We do not know with whom, but based on her calendar we can try to guess: she meets with her spouse 15% of the time; with her business partner 30% of the time; with her secretary 50% of the time; and with a secret visitor 5% of the time.

Sherlock Holmes is called in by the police. The woman has been shot! If only they knew whom she had her fatal meeting with. . . Fortunately, Sherlock Holmes took ORF 309 when he was an undergraduate at Princeton, so he knows what to do. First, he interviews the spouse, the business partner, and the secretary. Based on these interviews, Sherlock estimates that whenever the woman meets with her spouse, there is a 5% chance that he will get so upset that he shoots her (it seems there was a dispute over the laundry); when she meets with the business partner, there is a 3% chance that he pulls the trigger (it seems they have been having a disagreement about some mysteriously missing funds); and there is a 1% chance that she gets shot when she gets together with her secretary (who is a gun afficionado). Sherlock cannot interview the secret visitor, so he generously estimates a 5% chance that the secret visitor is an assassin. Which of these individuals is the most likely to have been the murderer?

Q. 3 (Help Chevalier de Méré)**.** The history of mathematical probability began when a French writer, Chevalier de Méré, could not figure out whether or not it is profitable to bet in the following game. A pair of dice is thrown 24 times (each die has six sides that are equally likely to come up, and the dice are thrown independently). Is it profitable to bet on the event that in these 24 throws, a double six is thrown at least once? De Méré was so confused that he had to ask his friend, the famous mathematician Blaise Pascal, for help. Help the poor Chevalier settle his question.

Q. 4 (Bluephobia)**.** Consider an urn containing 2 blue balls and 4 red balls. Due to your intense dislike of the color blue, you would like to get rid of the blue balls in the urn. To this end, you select a ball at random from the urn (each ball is equally likely to be chosen). If the ball is blue, you throw it in the trash; if it is red, you throw it back in the urn. You now repeat this process until there are only red balls left in the urn.

a. What is the probability that the process ends at the end of the fourth draw?

b. Given that the process ends at the end of the fourth draw, what is the probability that you get a blue ball in the third draw?

Q. 5 (Alice and Bob have fun with coins)**.** Alice and Bob are bored, so they decide to play the following game. Each of them independently flips a coin. If one of the outcomes is heads and one of the outcomes is tails, the person who got heads wins. Otherwise, they both flip their coins again, and repeat until one of them wins.

To make life interesting, both Alice and Bob flip a biased coin (one that is not equally likely to come up heads and tails). Alice's coin has probability $0 < p_A < 1$ of coming up heads, and Bob's coin has probability $0 < p_B < 1$ of coming up heads.

a. What is the probability that Alice wins the game in the first round?

b. What is the probability that Alice wins the game in the second round?

c. What is the probability that Alice wins the game in the kth round?

d. What is the probability that Alice eventually wins the game?

e. For what values of p_A and p_B is the game fair?